

does not give the total distance traveled as the object is not moving at constant speed.

$$\mathbf{9 \quad B \quad } a = \omega^2 r = \left(2\pi \times \frac{3000}{60} \right)^2 \left(\frac{8.0}{100} \right) = 7897 \approx 7900 \text{ m s}^{-2}$$

10. A. As the Earth is taken to be a uniform sphere, the gravitational potential at any point on the Earth's